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Seed-assisted polarity control of flexible WSe_2 transistors and demonstration of CMOS inverter

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Polarity engineering of transition metal dichalcogenide field-effect transistors (FETs) has become a key requirement for complementary logic. Here, we report the polarity control of flexible tungsten diselenide (WSe_2) FETs with atomic layer deposited (ALD) top-gate dielectrics. We identify evaporated WO_x as suitable seeds for ALD top-gates enabling p-channel WSe_2 FETs, while evaporated Al seeds lead to n-channel FETs. Our flexible p-channel FETs achieve good drain current on/off ratio up to 10^6, enhancement-mode operation with a negative threshold voltage, and low off-current down to ~6×10^{-7} μA/μm, which are important for low-power operation. Combining both seed layers in one fabrication process, we demonstrate flexible WSe_2 complementary metal-oxide-semiconductor (CMOS) inverters. Our inverters show good switching behavior with voltage gain up to 65 and total noise margin up to ~88%. Overall, this study provides a strategy on polarity engineering of flexible WSe_2 FETs and highlights the potential of flexible complementary WSe_2 electronics.

Two-dimensional (2D) transition metal dichalcogenides (TMDs) have gained increased interest for the next-generation flexible and wearable electronics due to their superior transistor scaling prospects, excellent electrical properties, and outstanding mechanical flexibility[1,2]. Among the various TMD semiconductors, tungsten diselenide (WSe_2) stands out as a particularly versatile channel material. In contrast to other TMDs (e.g., MoS_2) which typically exhibit unipolar n-type behavior, WSe_2 supports both n-type and p-type transports, making it highly suitable for complementary logic circuits using a single semiconductor[3]. WSe_2 also offers high carrier mobilities, up to 350 $cm^2V^{-1}s^{-1}$ for holes and 150 $cm^2V^{-1}s^{-1}$ for electrons at room temperature[3–5], potentially outperforming most flexible-compatible semiconductors such as organic materials[6], amorphous oxides[6,7], and low temperature polysilicon[8]. Its atomically thin nature suppresses short-channel effects more effectively than other materials, providing a promising route for device scaling, which could lead not only to notable performance improvements but also reduced power consumption[9,10]. These attributes highlight WSe_2 as a promising candidate for high-performance, low-power flexible complementary metal-oxide-semiconductor (CMOS) technologies[11].

To achieve truly flexible CMOS technology with WSe_2, it is important to control the polarity for n-channel and p-channel operation without degrading other field-effect transistor (FET) characteristics. Conventional doping strategies like in silicon are incompatible with the TMD crystal structure and substitutional doping has proven difficult due to deterioration of carrier mobility[12–14]. Thus, other means of doping have been explored, typically on rigid substrates, such as double gate engineering[15], contact engineering[16], interface charge control[17], and chemical doping[18]. Nevertheless, no prior work has established a reliable polarity-engineering strategy for WSe_2 implemented on flexible substrates. Related studies on other TMDs including MoS_2 and WS_2, have demonstrated that dielectric and interface engineering can effectively improve device performance. In particular, high k-dielectric integration has been shown to enhance electrostatic gate control, while interface and contact engineering strategies can reduce Schottky barrier height and contact resistance[19–21]. Furthermore, chemical doping has been reported to modulate carrier polarity of MoS_2 FETs[22]. These studies highlight the importance of dielectric, interface, and doping engineering for high-performance and polarity tuning of TMD electronics. However, polarity control is harder for top-gates than for back-gates and achieving such control in flexible devices is challenging due to several factors such as unintentional doping by top-gate dielectric deposition[23,24] and limited processing temperature range for flexible substrates[25]. Throughout the development of doping strategies, AlO_x has been reported as an n-type dopant for TMDs[24] and oxidized WSe_2 as a p-type dopant[26,27]. Evaporated WO_x has been used as an encapsulation layer[28], and adhesion contact layer for p-channel WSe_2 FETs on rigid substrates[29]. Besides that, ultra-thin (~1.5 nm) evaporated Al has widely been used as a seed layer for atomic

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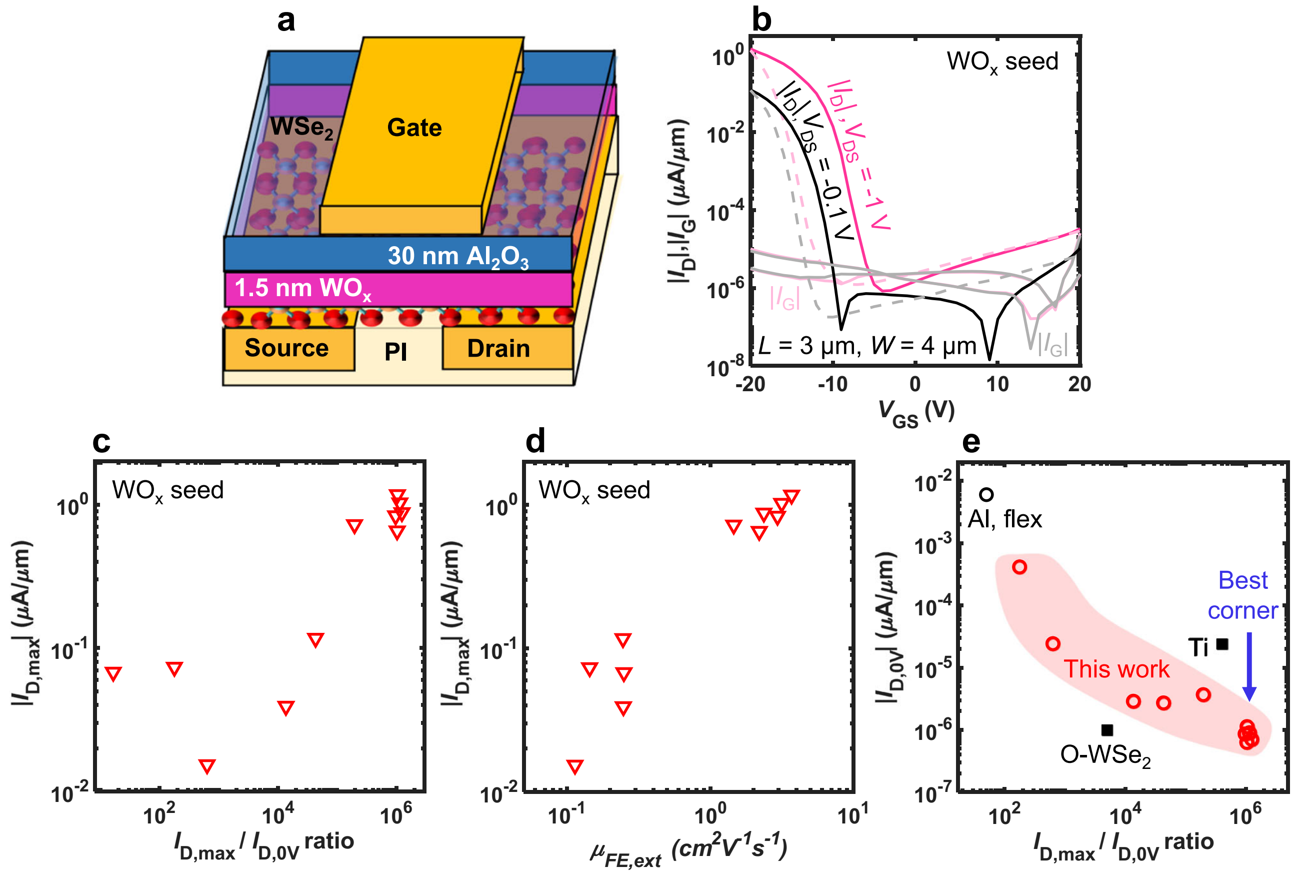

Fig. 1 | Flexible p-channel WSe_2 FETs with WO_x seed. **a** Schematic FET cross-section on polyimide (PI) substrate. **b** Transfer characteristics of a p-channel WSe_2 FET. The solid lines present the off-to-on V_{GS} sweeps and the dashed lines present the on-to-off V_{GS} sweeps. **c**, **d** $|I_{D,max}|$ extracted at $V_{GS} = -20$ V, I_D on/off ratio and $\mu_{FE,ext}$. The $\mu_{FE,ext}$ were extracted at $V_{DS} = -0.1$ V. **e** Benchmarking of off-current $|I_{D,0V}|$ vs. I_D on/off ratio of micron-scale top-gated p-channel WSe_2 FETs with different seeds (Al, Ti, oxidized WSe_2: O-WSe_2). Filled markers present devices on rigid substrates and unfilled markers present flexible devices. Best corner is the desired corner for low-power devices.

layer deposition (ALD), typically leading to more pronounced n-channel behavior of WSe_2 FETs due to the formation of AlO_x[30–32]. However, evaporated WO_x has not been investigated as a seed layer in top-gated devices nor in flexible FETs. We note that for top-gated devices, the seed between the dielectric and channel also acts as nucleation promoter facilitating the top-gate dielectric formation via thermal ALD[33].

In parallel with the development of complementary FETs, the study on CMOS inverters based on TMDs has also been reported. However, most existing demonstrations either rely on rigid substrates[17,34,35] or use different materials for the n- and p-channels (e.g., MoS_2 for n-channel, and WSe_2 for p-channel)[36,37]. The investigations into flexible CMOS inverters composed of TMD channel FETs remain very limited. Therefore, more studies are needed, especially toward using one semiconductor for both FET polarities, to realize the full potential of flexible TMD CMOS logic.

Overall, above mentioned works have separately provided approaches for WSe_2 polarity engineering via WO_x or via AlO_x deposition. However, there is no directly exchangeable process step established to alter polarity of WSe_2 FETs within a unified top-gate fabrication process, which is also applicable to flexible substrates. This has hindered the demonstration of flexible WSe_2 CMOS inverters.

In this work, we compare two evaporated seed layers (WO_x and Al) in flexible top-gated FETs using few-layer WSe_2. The ultrathin seed layers facilitate polarity control of flexible WSe_2 FETs which has potential for low-power enhancement-mode operation. The flexible p-channel WSe_2 FETs with WO_x seed reach good drain current I_D on/off ratio up to 10^6 and low off-current down to ~6×10^{-7} μA/μm. We also demonstrate flexible WSe_2 CMOS inverters, which show good switching behavior with voltage gain up to 65 and total noise margin up to ~88%. This approach is demonstrated on both few-layer exfoliated and chemical vapor deposited (CVD) WSe_2 which signifies the broader application of the method. Overall, our study provides a promising strategy on polarity engineering of WSe_2 directly on flexible substrates. It also fills a critical gap in the literature by demonstrating a flexible WSe_2 CMOS inverter using a seed-assisted top-gate dielectric.

Results and discussion

Flexible p-channel WSe_2 FETs with WO_x seed

For these devices, the WO_x seed layer was chosen to enable p-channel WSe_2 FETs. WSe_2 was exfoliated from bulk material with thicknesses ranging from 1.8 nm to 6.2 nm (Suppl. Fig. 1). Flexible top-gated FETs were fabricated in a process adapted from our previous works (Suppl. Fig. 2a)[32,38]. Figure 1a displays the schematic device cross-section. The devices are top-gated with staggered architecture. The dielectric stack consists of an evaporated 1.5 nm WO_x and 30 nm Al_2O_3 deposited by ALD. Raman spectra of WSe_2 on flexible polyimide (PI) before and after depositing the seed layer are shown in Suppl. Fig. 2b, indicating no detectable damage to the channel material.

Figure 1b displays the transfer characteristics of an exemplary device with gate-source voltage V_{GS} ranging from 20 V to −20 V at two different drain-source voltages V_{DS}. The device shows clear p-channel behavior with a maximum drain current $|I_{D,max}|$ ~1.2 μA/μm, a negative threshold voltage V_T ~ −9.8 V at $V_{DS} = -1$ V, and a low gate leakage current $|I_G|$. The device also shows good switching behavior with an I_D on/off ratio of ~10^6 (ratio of maximum drain current $I_{D,max}$ and off-current $I_{D,0V}$ at $V_{GS} = 0$ V). We note that the V_T has been extracted at a constant $|I_D| = 10$ nA/μm[39,40]. The

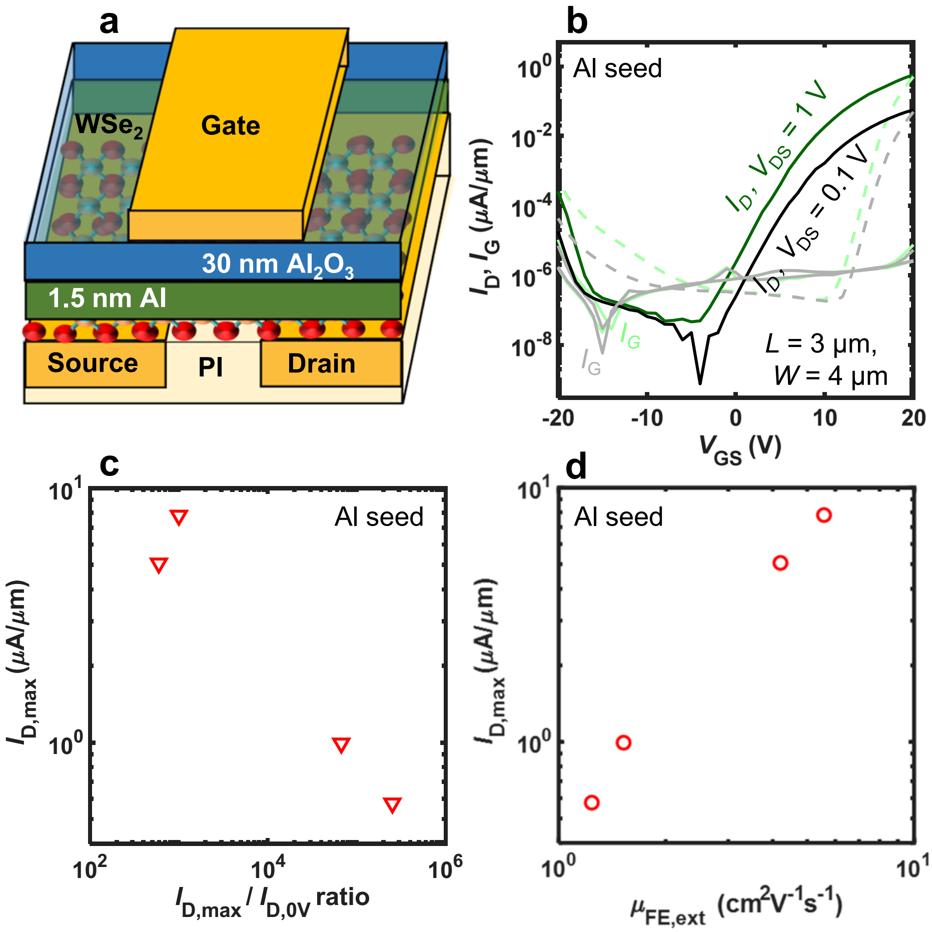

Fig. 2 | Flexible n-channel WSe_2 FETs with Al seed. a Schematic FET cross-section on polyimide (PI) substrate. **b** Transfer characteristics of an n-channel WSe_2 FET. The solid lines present the off-to-on V_{GS} sweeps and the dashed lines present the on-to-off V_{GS} sweeps. **c, d** Extraction of $I_{D,\ max}$ at $V_{GS} = 20$ V, I_D on/off ratio and $\mu_{FE,ext}$.

extracted $|I_{D,max}|$ vs. I_D on/off ratio of eleven devices are shown in Figure 1c. Figure 1d displays the extracted extrinsic field-effect mobility $\mu_{FE,ext}$ of eleven devices with a measured gate capacitance density of C_{ox} ~0.25 µF/cm^2 (Suppl. Fig. 3a). Assuming a thickness = 31.5 nm for the gate dielectric, we found a dielectric constant ε_r ~9 and an equivalent oxide thickness (EOT) is ~13.7 nm. Due to a high dielectric constant of WO_x (>20)[27,41,42], C_{ox} and ε_r of whole stack are higher compared to the Al seed stack (Suppl. Fig. 3b). The breakdown voltage V_{GS} is ~−25 V for the WO_x gate dielectric stack, indicating that the hybrid dielectric stack is of good quality (Suppl. Fig. 4a).

The p-doping of WSe_2 by WO_x is consistent with previous studies[43–45], which can be explained with the Fermi level alignment: WO_x has a high work function of up to ~6.4 eV[46], which is higher than the work function of few-layer WSe_2 (~4.5 eV)[47]. This difference leads to electron transfer from WSe_2 to WO_x at the interface to align the Fermi levels, resulting band bending in WSe_2 promoting its p-doping (Suppl. Fig. 5a)[48].

Next, we discuss the drain current $|I_{D,0V}|$ at $V_{GS} = 0$ V, which we define as off-current. We note that often the minimum I_D, typically at positive V_{GS}, is used as off-current, but for low-power enhancement mode operation, it is important to reach a low current level at $V_{GS} = 0$ V[49,50]. Engineering the channel/dielectric interface has also been used in other works to adjust V_T enabling low off-current[40,42,51], therefore we have compared our work to other thin interlayers between the ALD dielectric and the channel. Considering the goal of well-behaving enhancement-mode devices, we found the best results for I_D on/off ratio and off-current are ~10^9 and ~3×10^{-7} µA/µm, respectively, achieved with oxidized WSe_2 interlayer in a back-gated p-channel WSe_2 FET and demonstrated on a rigid substrate[42]. This supports that WO_x is a promising interlayer material for this purpose, however, it is generally more difficult to obtain good device performance using top-gates rather than back-gates. Figure 1e displays the benchmarking of off-current and I_D on/off ratio of micron-scale top-gated p-channel WSe_2 FETs with different interlayers, demonstrating the competitive performance of our devices[38,52,53]. Our FETs achieve I_D on/off ratio up to 10^6 and off-current down to ~6×10^{-7} µA/µm at $V_{DS} = -1$ V which are better than other top-gated WSe_2 FETs even with devices on rigid substrates, and is not far off from the best comparable back-gated WSe_2 FETs (more detailed data in Suppl. Table 1). The transfer characteristics of eleven devices and additional statistics of the I_D on/off ratio, off-current and V_T are shown in Suppl. Fig. 6.

Flexible n-channel WSe_2 FETs with Al seed

For these devices, the Al seed layer was chosen to obtain n-channel WSe_2 FETs. The thicknesses of exfoliated WSe_2 and fabrication process for n-channel FETs are similar as for p-channel FETs. The WSe_2 channel was also preserved without noticeable changes in the Raman peak after Al seed deposition (Suppl. Fig. 2c). We note that the thin 1.5 nm Al oxidizes in air forming AlO_x. The AlO_x is well-known for n-doping of WSe_2 FETs, which has been reported in literature[32,54–56]. Figure 2a displays the cross-section of the top-gated WSe_2 FETs with dielectric stack consisting of 1.5 nm Al and 30 nm Al_2O_3. Figure 2b shows the transfer characteristics of an exemplary device exhibiting n-channel behavior, with V_{GS} ranging from −20 V to 20 V. The device exhibits a good I_D on/off ratio of more than 10^5, V_T ~9 V, low off-current $I_{D,0V}$ ~2×10^{-6} µA/µm at $V_{DS} = 1$ V, and low I_G. Figure 2c displays the maximum drain current $I_{D,max}$ vs. I_D on/off ratio of four n-channel devices extracted at $V_{GS} = 20$ V. The transfer characteristics of the four devices are shown in Suppl. Fig. 7a. Our n-type FETs achieve $I_{D,max}$ up to ~7.8 µA/µm (at $V_{DS} = 1$ V) and $\mu_{FE,ext}$ up to 5.6 cm^{2}V^{-1}s^{-1} (extracted at $V_{DS} = 0.1$ V) (device transfer characteristics in Suppl. Fig. 7b). More extracted device data of n-channel FETs, fabricated within the inverter process presented below, can be found in Suppl. Fig. 7c, d. Additional statistics of off-current, I_D on/off ratio and V_T for ten devices are shown in Suppl. Fig. 7e–g. Figure 2d displays the extrinsic field-effect mobility $\mu_{FE,ext}$ of four devices with a measured gate capacitance density of C_{ox} ~0.23 μF/cm^2 (Suppl. Fig. 3b). Assuming a thickness of 31.5 nm for the gate dielectric, we found a dielectric constant ε_r ~8 and an EOT ~15.3 nm. The breakdown voltage is ~25 V for Al seed dielectric gate stack which is quite similar to the WO_x seed stack (Suppl. Fig. 4b). The AlO_x induced n-doping of TMDs has been frequently reported and can be attributed to positive fixed charges and/or interface charges[24,56,57]. A schematic band diagram of the AlO_x/WSe_2 interface illustrating n-type doping is shown in Suppl. Fig. 5b.

Comparing our n- and p-channel FETs, both dielectric stacks show good operation under the V_{GS} ranging from −20 V to 20 V, and become unstable or break at higher V_{GS} of ~± 25 V. The device statistics indicate noticeable variability (eg., I_D on/off ratio) for both WO_x and Al-seed FETs. This behavior can be associated with variation in WSe_2 thickness and surface quality, as well as possible local non-uniformity of the evaporated seed, which further influence the ALD process and contribute to the device variation. Nevertheless, the overall polarity trend remains consistent. The observed polarity is primarily supported by the electrical characteristics which are also consistent with previous literature. However, surface-potential or chemical-state measurements were not performed in the present study. Possible mechanisms, such as Fermi-level modulation, fixed charges and interface states at the channel and seed interface are therefore discussed as plausible mechanisms based on literature rather than direct verification, which should be verified in future interfacial material characterization studies. Besides that, the hysteresis of FETs with WO_x seed is relatively smaller compared to our devices with Al seed, indicating WO_x may form a better interface with WSe_2. The WO_x layer increases the physical distance between the Al_2O_3 defect bands and the channel (Suppl. Fig. 8). It may also passivate interface traps, leading to less charge trapping which reduces the hysteresis[42]. The analysis of hysteresis with different V_{GS} ranges, different V_{DS} and different sweep rates can be found in the Suppl. Fig. 9. Generally, hysteresis exhibits strong dependence on V_{GS} ranges, and less dependence on V_{DS} or sweep rate within the measurement timescale. This indicates that the hysteresis mainly originates from gate-field induced charge trapping. Notably, the hysteresis of both n- and p-channel FETs markedly reduce under pulsed measurements, indicating that slow charge trapping in border traps mainly contributes to hysteresis (Suppl. Fig. 9f, l)[58]. Overall, the evaporated WO_x seed provides promising interfacial behavior, which is similar or better compared to our reference Al-seed FETs. Future studies could investigate this aspect in more detail and explore approaches such as annealing, additional encapsulation, or optimization of the ALD process to minimize the hysteresis. In addition, investigating the influence of ALD deposition temperature within the thermal budget of PI (typically <300 ^{0}C, depending on the type of PI)[59] could provide further insight into the device characteristics including hysteresis, threshold voltage, and polarity. However, here we focus on the polarity control of top-gated WSe_2 FETs achieved with WO_x/Al seeds including a good suppression of the n-branch using WO_x.

We note that devices on such thin PI substrates are flexible and bendable. The thin substrate effectively minimizes the bending strain according to the equation strain = $d/(2\cdot r)$ where d is the thickness of substrate and r is the bending radius[32]. We have verified the mechanical stability of our devices at a bending radius of 4 mm corresponding to a strain <0.1%, which allows reliable probing on our probe station. We observed no notable changes in electrical characteristics for both the p-channel and n-channel devices under flat, bent and re-flattened conditions (Suppl. Fig. 10). We further note that we investigated the properties of flexible n-channel TMD transistors under strain in prior work and found that the mobility is enhanced under tensile strain, while changes were reversible and devices remained stable up to strain values of 0.7%[60]. With this in mind, we can estimate that the devices presented in this study, using substrate thicknesses of only 5 μm, could potentially allow for bending down to a radius of ~400 μm without notable damage. Due to the facility limitations, cyclic fatigue tests were not performed in this work. However, a previous study on flexible TMD FETs on 5 μm thick PI (same substrate thickness used in our work) has shown stable electrical characteristics over thousands of bending cycles in the millimeter-scale bending radius, suggesting the mechanical robustness of related flexible TMD device platforms[36].

Flexible CMOS inverters based on WSe_2

With the abovementioned strategy to control the FET polarity, we fabricated CMOS inverters based on WSe_2 on flexible PI substrates. This complementary circuit was achieved by integrating two WSe_2 FETs with the same metal contacts on the same substrate but using the two different seed layers. Figure 3a displays the fabrication process of the flexible inverters, which follows a similar process flow as the flexible FETs described above (see Methods section). However, for the inverter, the first seed layer WO_x was deposited as a blanket, after that a window was patterned for lift-off to define the area of the Al seed. We note that WO_x can be conveniently dissolved in AZ 726 MIF developer, therefore the WO_x layer in the window was removed during the development, before depositing Al.

Figure 3b shows the cross-sectional schematic of the flexible inverter based on WSe_2 channels with different seed layers. The input node is the common gate, and the output node is the shared contact of the two FETs. Figure 3c, d display the picture of the flexible PI chip and the optical image of the flexible WSe_2 inverter after fabrication.

Figure 3e displays exemplary transfer characteristics of the n- and p-channel exfoliated WSe_2 FETs used in the same inverter. Consistent with the above demonstration, the FET with WO_x exhibits p-channel characteristic and the FET with Al seed exhibits n-channel characteristic in the CMOS inverter integration. Both the n-channel and the p-channel devices have a channel length L = 3 μm, and a channel width W = 12 μm. Figure 3f shows the voltage transfer characteristic (VTC) of the flexible inverter, with output voltage V_{out} as a function of input voltage V_{in}, at supply voltage V_{DD} ranging from 2 V to 10 V. The VTC demonstrates good rail-to-rail behavior of an inverter with sharp voltage transition: the V_{out} remains at high level (close to V_{DD}) for low V_{in}, and switches to a low V_{out} for high V_{in}. Similar to a standard inverter, our device operates under positive input voltage. The switching threshold voltage (V_M) of the inverter representing the voltage when both n- and p-channel FETs are switched on with an equal $|I_D|$, has also a positive value. The VTC symmetry is not yet well balanced, thus the switching point does not occur at $V_{DD}/2$ which is due to the difference between electrical properties of the n-channel FET and the p-channel FET (e.g., $|I_D|$, V_T). As shown in the transfer characteristics of both FETs in Fig. 3e, the p-channel FET exhibits higher $|I_D|$ compared to the n-channel FET which pulls the output more to the V_{DD} causing V_M shifting to the right (positive) direction. Balancing the electrical properties of n- and p-channel FETs remains challenging, particularly for exfoliated WSe_2 where variations in flake thickness and interface quality may be present. However, this aspect can be optimized in future work e.g., for CVD material. As discussed below, we show that this method of achieving flexible WSe_2 CMOS electronics is also applicable to CVD material. The symmetry could potentially be improved by several approaches, including optimization of FETs dimensions (W/L ratios)[61], tuning the seed stoichiometry to adjust doping level[46], and contact engineering to tune carrier injection (source/drain) or the use of different gate metal work functions[16,62,63].The hysteresis of the inverter for double sweeps is still significant which requires further optimization in the future, possibly by annealing, more encapsulation, and a tuned ALD dielectric deposition process (Suppl. Fig. 11a)[64–66].

To characterize the switching behavior of our inverter, we extracted the voltage gain, which is defined as the derivative of the V_{out} with respect to the V_{in} and is an important parameter of the CMOS inverter. A voltage gain larger than unity is essential for cascaded logic operation, as it enables regenerative amplification and improves immunity to noise and signal errors[67]. Figure 3g displays the gain of the device for the forward sweep which is up to 65 at V_{DD} = 10 V. This good gain value is reflected in the steep voltage transition.

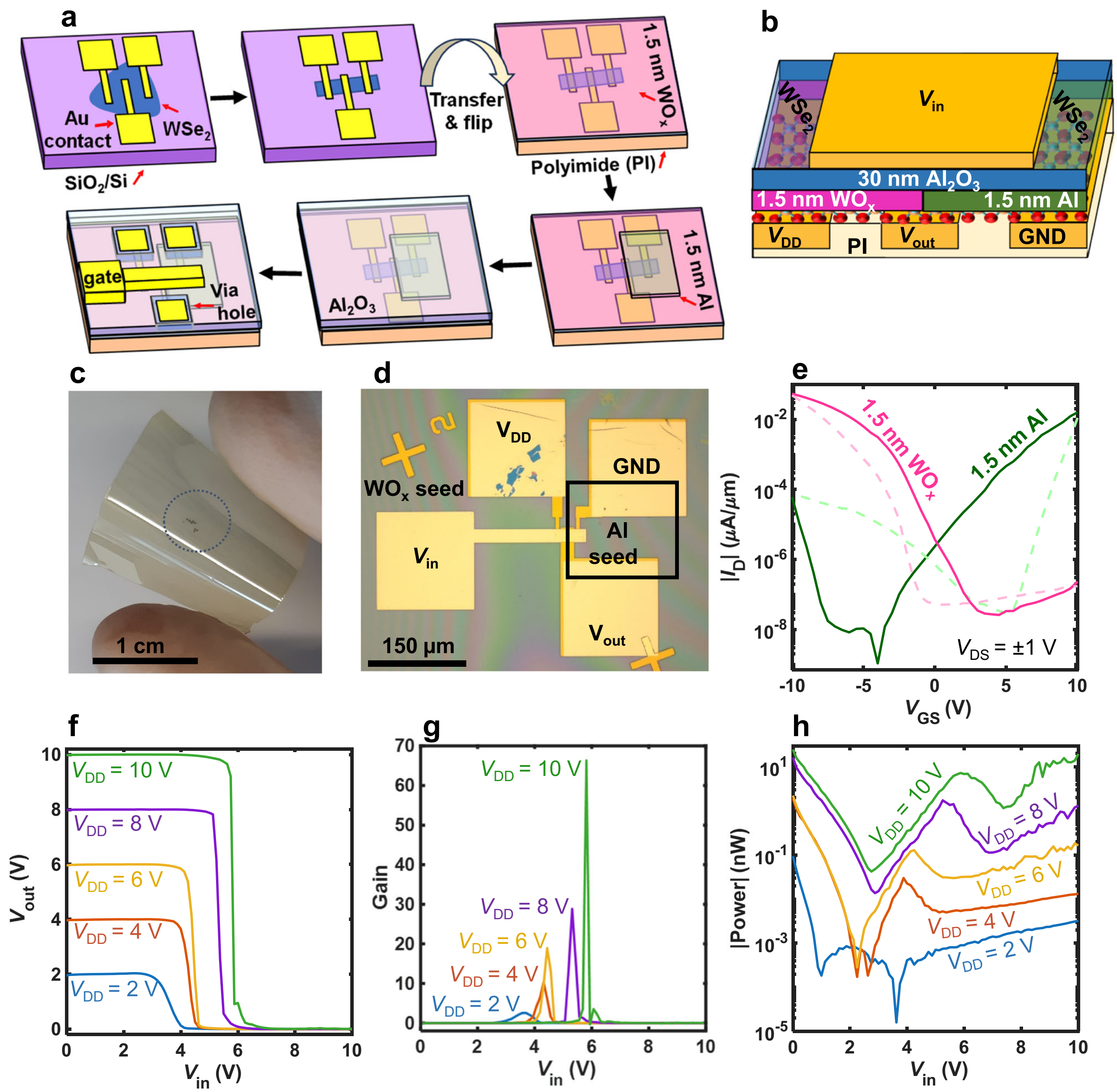

Fig. 3 | Flexible CMOS inverter based on WSe_2. a Integration process of flexible WSe_2 inverter. **b** Architecture of flexible WSe_2 inverter. **c** Picture of flexible chip after the fabrication (circle marks the area with devices). **d** Optical image of flexible inverter. **e** Transfer characteristics of the two FETs in the inverter. The solid lines present the off-to-on V_{GS} sweeps and the dashed lines present the on-to-off V_{GS} sweeps. **f** Voltage transfer characteristics of the flexible inverter. **g** Gain extraction of flexible inverter. **h** Static power consumption of flexible inverter. Data in (**e**–**h**) are from the same inverter and displayed for different supply voltages V_{DD}. The data in (**f**–**h**) of the inverter are for the forward sweep direction.

Figure 3h displays the static power consumption as function of V_{in}, following the equation $P = V_{DD} \times I_{DD}$, where I_{DD} is the current through the inverter. The peak of power consumption of the inverter is ~25 nW at $V_{DD} = 10$ V. However, the power peak reaches 95 pW when $V_{DD} = 2$ V demonstrating the promise of our approach for WSe_2 CMOS technology toward low-power flexible electronics. The static current I_{DD} vs. V_{in} of this device is shown in Suppl. Fig. 11b. At high V_{DD}, I_{DD} shows a notable increase at low- and high-V_{in} levels, which can be attributed to non-ideal off-current, residual ambipolar conduction, FETs mismatch and hysteresis in both n- and p-channel FETs. This can be improved by several approaches such as optimizing the symmetry of the transistors[18], the overlap between the gate and contacts[68], and reduction of EOT[69]. We note that this work mainly focuses on static inverter operation based on flexible WSe_2 FETs. As a first-order estimate, the intrinsic switching speed can be approximated from the resistive-capacitive (RC) charging delay $t_d \approx C_L V_{DD}/I_{on}$, where C_L is load capacitance and I_{on} is on-state current of the FET[67]. Using our measured gate capacitance density (calculated above for each type of FET) and inverter geometry for C_L, we could estimate an intrinsic operating frequency ranging in the order of 100 kHz for our devices. The practical operating speed is still limited by channel mobility, parasitic overlap capacitance, and possibly contact resistance, which were not optimized in this study[70,71].

Next, we discuss the noise margin, which is the amount of noise that the device could withstand without any unwanted switching. It was extracted at unity voltage gain to evaluate the robustness of the inverter. The noise margin high is defined as $NM_H = V_{OH} - V_{IH}$, where V_{OH} and V_{IH} are the highest output and input voltages, respectively. The noise margin low is

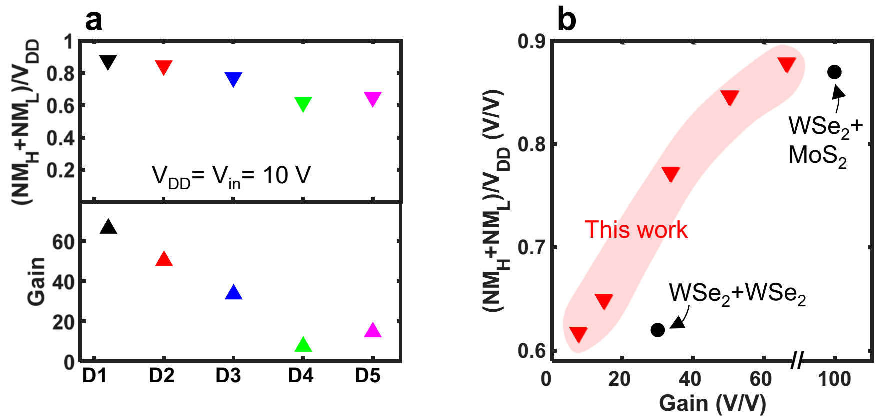

Fig. 4 | Statistics of flexible WSe_2 inverters and benchmarking. **a** Total noise margin and gain extraction of five inverters. **b** Benchmarking of flexible inverters based on TMDs (the inverter in the other work with WSe_2/WSe_2 channels used an ion gel based-dielectric). Our data were extracted from forward sweeps.

defined as $NM_L = V_{IL}$-V_{OL} where V_{IL} and V_{OL} are the lowest input and output voltages, respectively. The total noise margin equals $NM_H + NM_L$. Our device achieves good total noise margin up to ~88%, demonstrating strong noise tolerance (Suppl. Fig. 11c). We note that this only considers the forward sweep and hysteresis needs to be minimized to fully leverage this potential. Figure 4a displays the total noise margin and gain extraction of five flexible inverters. Our devices show good total noise margin with an average value ~75% and an average gain ~33 extracted from forward sweeps. The inverter-to-inverter variation is consistent with the device-to-device variation observed in the separately characterized n- and p-channel FETs. Because each inverter consists of one n-channel and one p-channel WSe_2 FET, variations in threshold voltage and current level of each FETs directly affect the performance of the inverter. Details of device data can be found in Suppl. Fig. 12. The VTC of an inverter under repeated voltage switching is shown in Suppl. Fig. 13a indicating the stability of our device. We note that despite hysteresis, the total noise margin and gain are quite similar for both forward and reverse sweeps (Suppl. Fig. 13b).

Given the high potential of TMDs in CMOS technology, numerous studies have demonstrated their application in CMOS inverters. To the best of our knowledge, a CMOS inverter solely based on WSe_2 on a rigid substrate has been reported with a highest voltage gain ~340 and a total noise margin ~90%, which uses different source/drain contact integration methods for the n- and p-channel FETs (evaporated Au contacts and transferred Au contacts)[72]. Others also reported good CMOS inverters based on hybrid channels from MoS_2/WSe_2 inks on rigid substrates with a voltage gain ~460 and a total noise margin ~90%[37]. So far, the reports on flexible CMOS inverters based on TMDs still remain very limited. Pu et al. presented the first flexible CMOS inverter based on WSe_2 for both n- and p-channels but used an ion gel as a gate dielectric, reporting a gain ~30 and a total noise margin ~62%[73]. Piacentini et al. demonstrated a flexible CMOS inverter in a back-gate configuration, based on hybrid channel materials (MoS_2 and WSe_2) with a gain ~30–100 (depending on the V_{in} sweep step size) and a total noise margin ~87%[36]. Zou et al. reported flexible inverters based on MoS_2/WSe_2 inks with gain up to ~61 at $V_{DD} = 9$ V[37]. The benchmarking of flexible CMOS inverters based on TMDs, including gain and noise margin, is presented in Fig. 4b. This reveals the scarcity of reports on CMOS inverters with flexible TMD transistors (details in Suppl. Table S2). Our devices show voltage gain up to 65 and total noise margin up to ~88%, which is competitive to other works. The balance between NM_H and NM_L of our inverters is also comparable to others (Suppl. Fig. 13c).

With the same approach, we also demonstrated polarity control of FETs with few-layer CVD WSe_2 and realized flexible inverters (Suppl. Fig. 14). We obtained good switching behavior and achieved a gain of ~15 at $V_{DD} = 5$ V. We note that the inverter with the CVD WSe_2 was not stable under higher V_{DD} (e.g., 10 V) which can be due to the WSe_2 material properties and dielectric quality. Further improvement can be achieved by optimization of CVD conditions, seed deposition, or post-growth process steps. Overall, these results demonstrate a promising step toward flexible WSe_2 CMOS electronics using seed layers for top-gates with solid-state dielectrics.

In summary, flexible p-channel WSe_2 FETs enabled by evaporated WO_x seed layers for ALD top-gates have been presented. This seed layer revealed promising doping effectiveness, leading to unipolar p-channel behavior with good I_D on/off ratio up to 10^6 and low off-current down to 6×10^{-7} μA/μm. Other device properties like the $|I_{D,max}|$, I_D on/off ratio, hysteresis, gate leakage, and gate capacitance were similar or better than for our reference n-channel FETs with Al seeds, the common approach for top-gated TMD FETs. By utilizing the two different seeds, our work demonstrated the tuning of polarity of flexible WSe_2 FETs compatible with top-gated dielectrics. Furthermore, we also combined both Al and WO_x seeds in one fabrication process and created flexible WSe_2 CMOS inverters. Our inverters show good switching characteristics with voltage gain up to 65 and total noise margin up to ~88%. Compared to previous ion gel dielectrics or hybrid channel integration, this represents a promising route for flexible WSe_2 CMOS electronics based on solid-state gate dielectrics. Although our initial result revealed that this fabrication process needs further improvement in the future (e.g., EOT scaling and minimization of hysteresis), we demonstrated effective polarity control and CMOS functionality using the evaporated seed layers. Overall, this work presents a promising step toward flexible CMOS electronics solely with WSe_2 as a channel material.

Methods

Material

Few-layer WSe_2 (1.8 nm to 6.2 nm) were prepared by mechanical exfoliation from a bulk crystal on Si/SiO_2 substrates (bulk flux-grown WSe_2 was purchased from 2D materials). First, the substrates were cleaned in ozone plasma for 15 minutes (power density of 50 mW/cm^2 and flow rate of 500 sccm O_2) to remove residues and create a hydrophilic surface for better adhesion with WSe_2. Bulk WSe_2 was first cleaved using scotch tape (3M Magic) and repeatedly folded and peeled to thin the material. The thin layers were then transferred on the Si/SiO_2 substrate by pressing the tape onto the substrate surface and slowly removing it. After transfer, the few-layer WSe_2 was inspected using an optical microscope. The actual thickness of WSe_2 was measured using atomic force microscopy (AFM). Most devices in this manuscript have been fabricated with exfoliated WSe_2, but we have also verified that similar results can be achieved with few-layer CVD WSe_2 (see Suppl. Fig. 1 and Suppl. Fig. 14).

For the CVD material, WSe_2 was synthesized using Se powder as the chalcogen source and a mixture of NaCl and $WO_{2.9}$ as both the growth promoter and tungsten precursor, respectively. The precursor mixture was positioned at the center of the furnace, while the Se powder was placed upstream to facilitate vapor transport. A SiO_2/Si substrate was placed facedown directly above the tungsten precursor. The growth was conducted at 825 °C for 15 min under a controlled flow of Ar/H_2 (50/10 sccm), followed by natural cooling to room temperature[74,75].

Flexible top-gated FET fabrication

Flexible top-gated FETs were fabricated in a similar process as demonstrated in our previous works (Figure S2a)[32,38]. Initially, 50 nm Au source/drain contacts were electron-beam evaporated and patterned by lifted-off. The channel was patterned using a photoresist (PR) mask and etched using O_2/CF_4 plasma at gas flows of 10 sccm/30 sccm, 50 W power and ~7 mTorr pressure (PlasmaTherm RIE). The PR was removed in solvent stripper (Technistrip NI555, acetone and iso-propanol). Next, a 5 μm thick polyimide (U-Vanish S, UBE Europe GmbH) layer was spin-coated on top, baked at 120 °C for 2 min and cured in N_2 at 250 °C for 30 min. The PI/Au/WSe_2 stack was transferred in deionized water. Subsequently, a 1.5 nm thick seed layer (WO_x or Al) was electron-beam evaporated at a rate of 0.1–0.2 Å/s, followed by the deposition of 30 nm Al_2O_3 using thermal ALD at 200 °C (trimethylaluminum (TMA) precursor). Finally, a 50 nm Au top-gate was deposited by electron-beam evaporation and lifted-off, and the contacts were opened by wet etching of Al_2O_3 in standard Al etchant at 40 °C (TechniEtch Al80). During fabrication steps (e.g., lithography), the PI film was temporarily attached to a rigid Si substrate using PR to provide sufficient mechanical stability for processing[76]. For lift-off processes, the PI chip was immersed in lift-off solvent for several hours until the unwanted metal was completely detached.

Characterizations

Electrical measurements of FETs and inverters were performed in ambient conditions using Keysight B1500A Semiconductor Device Parameter Analyzer. The thicknesses of WSe_2 were measured using AC mode imaging (JPK NanoWizard4 AFM system). Raman measurements were performed using an excitation laser of 532 nm, laser power 1 mW (WITec Alpha300R Raman microscope). Extrinsic field-effect mobility $\mu_{\mathrm{FE,ext}}$ was extracted at $V_{\mathrm{DS}} = \pm 0.1$ V following the equation of maximum transconductance $g_m = \mu_{\mathrm{FE,ext}} C_{\mathrm{ox}} V_{\mathrm{DS}} W/L$.

Data availability

Relevant data supporting the key findings of this study are available within the article and the Supplementary Information file. All raw data generated during the study are available from the corresponding author upon reasonable request.

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Author contributions

Q.T.P. performed the device fabrication, characterization, and analysis under the supervision of A.D. K.M.N. performed the CVD of WSe_2 and contributed to CVD material characterization. J.Y.P. supervised the CVD of WSe_2. A.D. supervised the overall study. A.D. and Q.T.P. wrote the manuscript. All authors read and approved the manuscript.

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